

APPENDIX B Classification of Life

This appendix presents a taxonomic classification for the major extant groups of organisms discussed in this text; not all phyla are included. The classification presented here is based on the three-domain system, which assigns the two major groups of prokaryotes, bacteria and archaea, to separate domains (with eukaryotes making up the third domain).

Various alternative classification schemes are discussed in Unit Five of the text. The taxonomic turmoil includes debates about the number and boundaries of kingdoms and about the alignment of the Linnaean classification hierarchy with the findings of modern cladistic analysis.

DOMAIN BACTERIA

- **Proteobacteria**
- **Chlamydia**
- **Spirochetes**
- **Cyanobacteria**
- **Gram-positive Bacteria**



DOMAIN ARCHAEA

- **Euryarchaeota**
- **Thaumarchaeota**
- **Aigarchaeota**
- **Crenarchaeota**
- **Korarchaeota**



DOMAIN EUKARYA

In the phylogenetic hypothesis we present in Chapter 28, major clades of eukaryotes are grouped together in the four “supergroups” listed in blue type. Formerly, all the eukaryotes generally called protists were assigned to a single kingdom, Protista. However, advances in systematics have made it clear that some protists are more closely related to plants, fungi, or animals than they are to other protists. As a result, the kingdom Protista has been abandoned.

Excavata

- Diplomonadida (diplomonads)
- Parabasala (parabasalids)
- Euglenozoa (euglenozoans)
 - Kinetoplastida (kinetoplastids)
 - Euglenophyta (euglenids)

SAR

- Stramenopila (stramenopiles)
 - Oomycota (oomycetes)
 - Phaeophyta (brown algae)
 - Bacillariophyta (diatoms)



- Alveolata (alveolates)
 - Dinoflagellata (dinoflagellates)
 - Apicomplexa (apicomplexans)
 - Ciliophora (ciliates)
- Rhizaria (rhizarians)
 - Radiolaria (radiolarians)
 - Foraminifera (forams)
 - Cercozoa (cercozoans)

Archaeplastida

- Rhodophyta (red algae)
 - Chlorophyta (green algae: chlorophytes)
 - Charophyta (green algae: charophytes)
 - Plantae
 - Phylum Hepatophyta (liverworts)
 - Phylum Bryophyta (mosses)
 - Phylum Anthocerophyta (hornworts)
 - Phylum Lycopphyta (lycophytes)
 - Phylum Monilophyta (ferns, horsetails, whisk ferns)
 - Phylum Ginkgophyta (ginkgo)
 - Phylum Cycadophyta (cycads)
 - Phylum Gnetophyta (gnetophytes)
 - Phylum Coniferophyta (conifers)
 - Phylum Anthophyta (flowering plants)
- Nonvascular plants (bryophytes)
- Seedless vascular plants
- Gymnosperms
- Angiosperms
- Seed plants

